

Affordable, Portable, Real-Time Chlorophyll Sensor

Physics Report | Group 04

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1 Executive Summary

This report details the development of a portable, high-accuracy chlorophyll sensor utilizing **Laser-Induced Fluorescence (LIF)** and **tinyML** algorithms. Designed for agricultural, herbal, and aquatic monitoring, the system provides real-time, calibrated measurements of both Chlorophyll-a (Chl-a) and Chlorophyll-b (Chl-b) concentrations.

2 Scientific Foundations

2.1 Importance of Chlorophyll

Chlorophyll is a primary indicator of plant health, product quality in the herbal industry, and algal biomass in aquatic ecosystems. Monitoring its concentration is critical for water quality assessment and precision agriculture.

2.2 Detection Methodology

While traditional methods (HPLC, Mass Spectrometry) offer high precision, they are expensive and laboratory-bound. Existing affordable meters often provide only relative content and lack absolute calibration.

- **Research Gap:** Creating an affordable sensor providing calibrated, simultaneous measurements of Chl-a and Chl-b.
- **Objective:** Prototype a real-time sensor using LIF and on-device edge computing analysis.

3 Optical & System Methodology

3.1 Fluorescence Physics

The sensor employs a flow-through system where samples are excited by targeted wavelengths: * **Laser 405 nm:** Optimized for Chl-a excitation. * **Laser 450 nm:** Optimized for Chl-b excitation.

The resulting fluorescence is captured by a CCD sensor for spectral analysis.

3.2 System Architecture

- **Control Unit:** STM32H743VIT6 (480MHz clock) for high-speed acquisition and on-device inference.
- **Sensor:** Toshiba TCD1304 linear CCD (3694 pixels).
- **PCB Design:** A custom PCB was developed using **KiCad** and manufactured via **JLCPCB** to ensure signal integrity and portability.

4 Signal Processing & tinyML

4.1 Preprocessing Pipeline

To mitigate noise and baseline drift, the signal undergoes several mathematical steps:

$$\text{Clean Signal} = (65535 - \text{ADC Value}) - \text{Dark Signal}$$

1. **Inversion & Dark Subtraction:** Converting inverted CCD output and removing dark current.
2. **Median Filtering:** Removing single-pixel spikes.
3. **Savitzky-Golay (SG):** Smoothing and 1st derivative calculation to identify spectral slope.
4. **Standard Normal Variate (SNV):** Normalizing intensity for zero mean and unit variance.

4.2 Machine Learning Models

Built with **Edge Impulse**, the models run directly on the STM32: * **Chl-a:** Partial Least Squares (PLS) Regression. * **Chl-b:** Support Vector Regression (SVR) with RBF Kernel.

5 Experimental Results

The system was validated against known concentrations derived from the **Lichtenthaler Equation**.

Analyte	Algorithm	Train R^2	Test R^2	RMSE (mg/L)
Chl-a	PLS	0.992	0.867	~1.55
Chl-b	SVR (RBF)	0.992	0.842	~1.07

5.1 Uncertainty Analysis

Monte Carlo simulations confirm that uncertainty increases at higher concentrations, likely due to: * **Inner Filter Effect (IFE)**: Re-absorption of fluorescence at high densities. * **Fluorescence Saturation**: Non-linear response above 18 mg/L.

6 Project Economics (Budget)

The system is designed for maximum affordability without compromising spectral precision. * **Optical Components**: Lasers (405nm, 450nm), Diffraction Grating. * **Electronics**: STM32H7 MCU, TCD1304 CCD, LCD. * **Manufacturing**: Custom PCB (KiCad/JLCPCB). * **Infrastructure**: 3D Printed housing, flow-through sample chamber.

7 Limitations & Future Work

7.1 Limitations

- **Concentration Drift**: Acetone evaporation and chlorophyll degradation during testing.
- **Instrumental Noise**: CCD electronic noise and sensitivity limits.

7.2 Improvements

- **Upgraded Optics**: 1200 lines/mm diffraction grating and objective lens for beam collimation.
- **Drift Mitigation**: Sealed sample chambers to prevent evaporation.

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9 References

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